# BLE Integration & Web App Communication Summary

ECE Bio-Metric Wearable Sensor Project

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## 1 Executive Summary

This document summarizes the successful integration of the MAX30102 optical pulse oximeter algorithm with a Bluetooth Low Energy (BLE) GATT server on the Raspberry Pi Zero W. The system transitions from local terminal-only logging to wirelessly streaming continuous Heart Rate (BPM) and Blood Oxygen ($SpO_2$) metrics directly to a Chrome Web Bluetooth client interface (**Uplift HR**).

## 2 System Architecture & Communication Protocol

### 2.1 Peripheral / Central Hierarchy

- **BLE Peripheral (GATT Server):** Raspberry Pi Zero W running Linux `BlueZ` via Python's `bluezero` framework.
- **BLE Central (GATT Client):** Web Bluetooth application running on Google Chrome (`index.html` / `app.js`).
- **Device Advertisement Name:** `Uplift HR`

### 2.2 GATT Service & Characteristic Specification

To ensure compatibility with standard health platforms and custom web interfaces, the Pi advertises using standard Bluetooth SIG defined UUIDs:

| Entity | UUID | Properties |
|---|---|---|
| Heart Rate Service | `0x180D` (`0000180d-0000-1000-8000-00805f9b34fb`) | Primary Service |
| Heart Rate Measurement | `0x2A37` (`00002a37-0000-1000-8000-00805f9b34fb`) | Read, Notify |

Table 1: GATT Service and Characteristic Configuration

### 2.3 Payload Data Structure

Heart rate data is formatted into standard 2-byte BLE notification payloads:

- **Byte 0 (Flags):** `0x00` (Specifies 8-bit unsigned integer BPM format).
- **Byte 1 (Value):** Unsigned integer representing instantaneous smoothed Heart Rate ($0$–$255$ BPM).

## 3 Prerequisites & Package Setup

To establish the BLE GATT server on Raspberry Pi OS, essential system packages and Bluetooth stack dependencies must be installed globally for the root environment:

```
sudo apt update
sudo apt install -y python3-pip bluez python3-dbus python3-gi
sudo pip3 install bluezero --break-system-packages
sudo rfkill unblock bluetooth
```

- **`bluez` & `python3-dbus`:** Provides low-level Linux Bluetooth daemon protocols and D-Bus IPC interface wrappers.
- **`bluezero`:** Provides a Python abstraction layer over BlueZ D-Bus APIs for peripheral advertising and GATT service creation.
- **`sudo pip3 ... --break-system-packages`:** Ensures `bluezero` is accessible to the root user environment required when launching hardware scripts with `sudo`.
- **`rfkill`:** Soft-unblocks the onboard Bluetooth hardware interface on the Raspberry Pi Zero W.

## 4 Software Implementation Details

The production script `uplift_hr_pi.py` implements a multi-threaded architecture:

1. **Background Bio-Metric Thread (`sensor_worker`):**
   - Continuous $100$ Hz sampling from the MAX30102 sensor via $\mathrm{I}^2\mathrm{C}$ (Address `0x57`).
   - Sliding window buffer management ($300$ samples / $3.0$ seconds).
   - Real-time running mean DC trend removal and 5-point low-pass smoothing.
   - Adaptive statistical peak detection (Mean$+0.4\times$StdDev) with physiological bound checks ($35$–$180$ BPM).
   - Exponential Moving Average filtering ($\alpha = 0.3$) for output display stability.
2. **Asynchronous BLE Event Loop (`bluezero`):**
   - Publishes GATT server characteristics and handles connection events.
   - Dispatches periodic BLE notifications to subscribers every $1.0$ second.

## 5 Debugging & Issues Resolved

During implementation, three technical obstacles were identified and resolved:

| Issue | Root Cause | Resolution |
|---|---|---|
| `TypeError: unexpected keyword argument 'name'` | Parameter mismatch in `bluezero.peripheral.Peripheral` API. | Updated initialization parameter from `name` to `local_name`. |
| Adapter MAC Binding Error | Hardcoded Bluetooth adapter address caused binding failures across different boards. | Implemented dynamic adapter discovery via `adapter.Adapter.available()`. |
| `ModuleNotFoundError: No module named 'bluezero'` | Running with `sudo` isolates python environments from user-installed packages. | Re-installed dependencies globally via `sudo pip3 install bluezero --break-system-packages`. |

Table 2: Debugging Summary

## 6 Verification & Test Results

1. **BLE Advertisement:** On launch, the Pi successfully broadcasts as `Uplift HR` over the onboard Bluetooth controller.
2. **Web App Pairing:** Chrome's Web Bluetooth interface successfully discovers, pairs with, and subscribes to notifications from `0x2A37`.
3. **Data Streaming:** Verified real-time telemetry output in terminal:

   ```
   Sent 72 BPM | SpO2: 98% | Status: OK
   ```

4. **Finger Removal Graceful Handling:** When finger contact is broken, state machine correctly drops values to $0$ BPM and emits `Status: No finger detected` without dropping the BLE connection.

## 7 Next Steps

- Deploy a system background daemon (`systemd` service `biometric.service`) to execute `uplift_hr_pi.py` automatically on Pi power-up (headless operation).
- Proceed with Phase 1 hardware refinement (installing $1.8$V LDO regulator and active $\overline{\text{INT}}$ pin integration).